

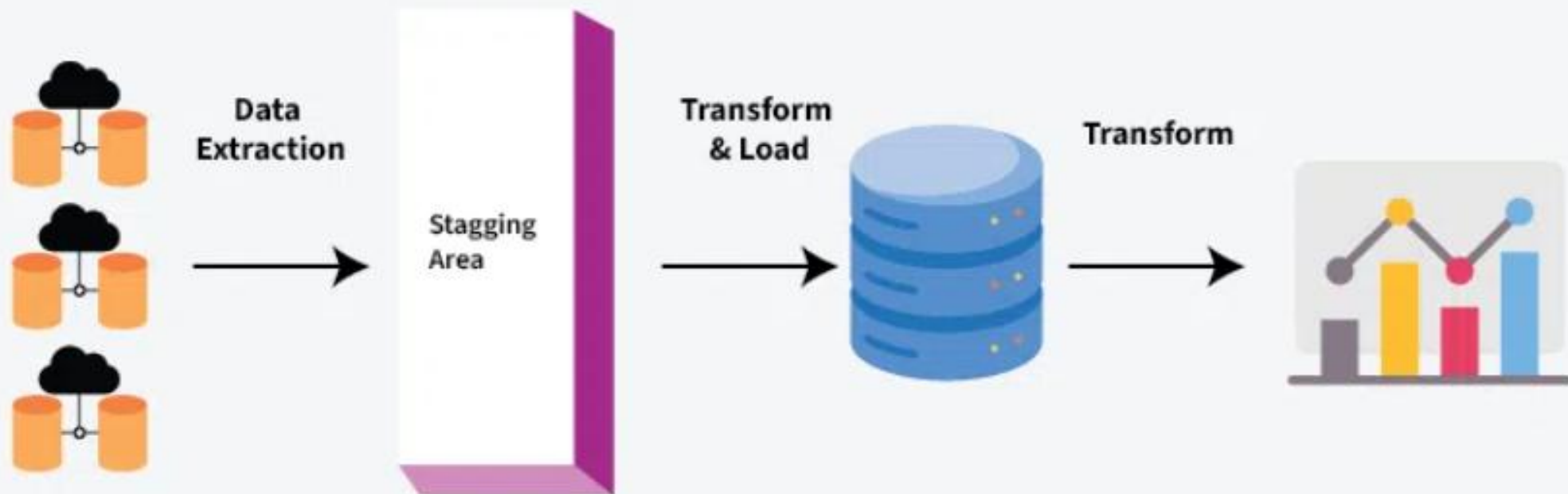
Data Pipeline

Lecture 2

Overview of Data Pipeline

- A data pipeline is a set of tools and processes for collecting, processing, and delivering data from one or more sources to a destination where it can be analyzed and used.
- Data Pipeline deals with information that is flowing from one end to another. In simple words, we can say collecting the data from various resources than processing it as per requirement and transferring it to the destination by following some sequential activities.
- It is a set of manner that first extracts data from various resources and transforms it to a destination means it processes it as well as moves it from one system to another system.
- A well-designed data pipeline is crucial for organizations to leverage their data effectively, support decision-making, and gain insights that drive business success

Data Pipeline Overview



Why Data Pipelines are important?

- Let's think about a scenario where a data pipeline is helpful.
- The improvement of the cloud has meant that modern technology for enterprises uses lots of apps with different features. The retailing team might employ a combination of Hub spot and Market for trading automation.
- The other retailer teams mostly depend on Salesforce to handle and some might use MongoDB for storing customer approaches.
- This leads to the waste of data across different tools and results in data silos. Data silos are nothing but they will create it difficult to fetch even business insights, like your most profitable market.
- It is most important for Business Intelligence(BI) in their day-to-day life they require everyday information to work with.

How to build a Data Pipeline?

- An organization can decide the methods of development to be followed just to abstract data from sources and transfer it to the destination.
- Batch transforming and processing are two common methods of development. Then there is a decision on what transformation process- ELT(Extract/Load/Transform) or ETL -to use before the data is moved to the required destination.

Challenges to building Data Pipeline

- Netflix, has built its own data pipeline. However, building your own data pipeline is very difficult and time is taken.

Here are some common challenges to creating a data pipeline in-house:

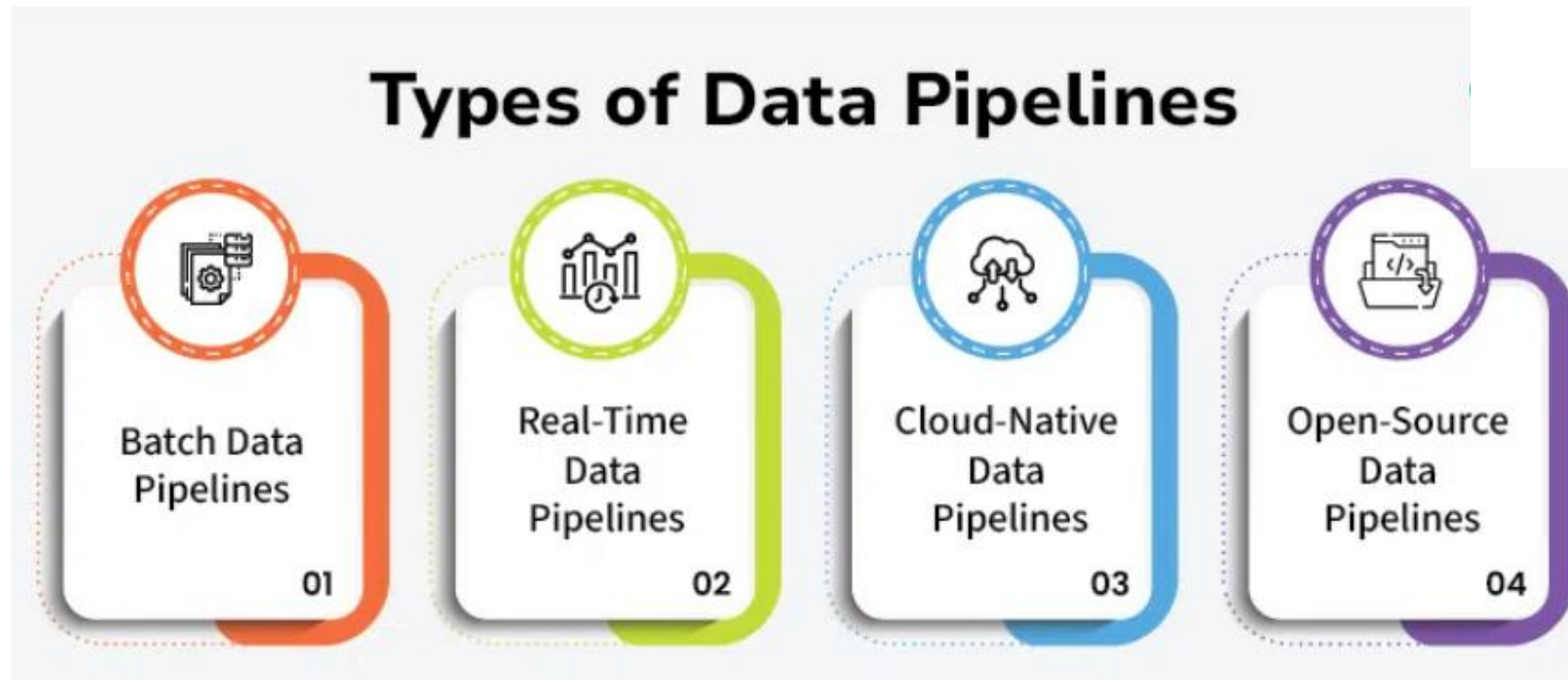
- Connection
- Flexibility
- Centralization
- Latency

Components of Data Pipeline :

- To know deep about how a data pipeline prepares large datasets for deconstruction, we have to know it is the main component of a common data pipeline. These are –
- Source
- Destination
- Data flow
- Processing
- Workflow
- Monitoring

Types of Data Pipelines

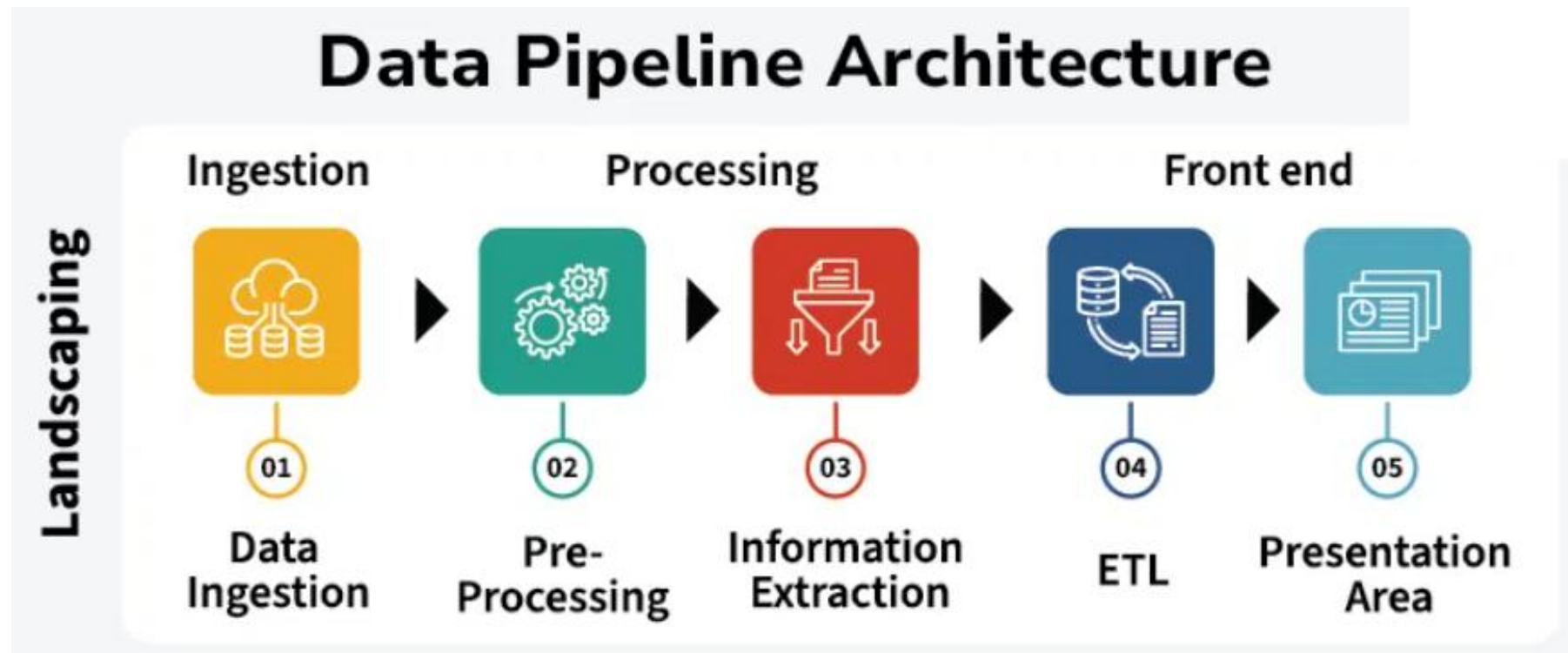
- Here are the following types of data pipeline:



- **Batch Data Pipelines:** Interact with large portions of data all at once and at some specific time of the day.
- **Real-Time Data Pipelines:** Interact with data at the time of its creation for almost real-time outcome.
- **Cloud-Native Data Pipelines:** Built for running in cloud environments which are more malleable and more flexible.
- **Open-Source Data Pipelines:** Created with the implementation of the open-source technologies like Apache Kafka, Airflow or Spark.

Data Pipeline Architecture

- Here is the representation of data Pipeline Architecture:



- **Ingestion Layer:** It retrieves data from an assortment of sources ranging from databases to APIs or even event streams.
- **Processing Layer:** Another operation on data involves the analysis of the said data followed by data cleaning through tools such as spark or Hadoop.
- **Storage Layer:** Held in data lakes, warehouses or other databases, data was kept for future reference or as a back up to be analyzed later.
- **Monitoring Layer:** It is the authority that is charged with the responsibility of providing quality data in the right time and increasing the efficiency of the system.
- **Consumption Layer:** Delivers the final data to BI tools or machine learning models where the data is analyzed at the next level for decision making.

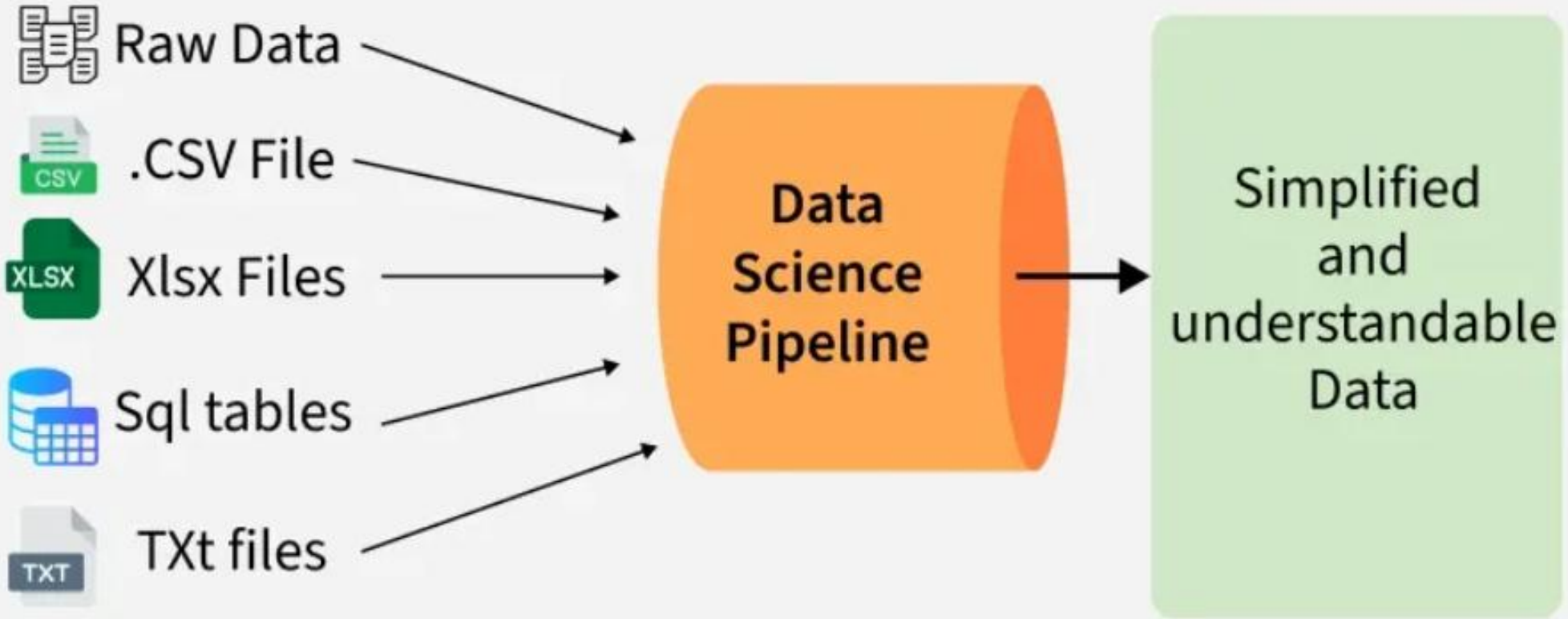
- In summary, while both data pipelines and ETL pipelines are used to manage and process data, ETL pipelines are a specific type of data pipeline with a focus on the Extract, Transform, and Load stages. Data pipelines have a broader scope and can include additional processing and delivery stages.

- **Some of the use cases of data pipelines:**
- **Real-Time Analytics:** Benefits real-time analytics and operational decisions in areas such as finance or healthcare among others.
- **Machine Learning:** Transmits data obtained after analysis from multiple feeds to the ML models for analysis and automation.
- **Data Migration:** Move data from one system to another especially when one has been enhanced or in the process of migrating to cloud environment.
- **Data Integration:** Combines two or more databases to look at the data with one common database.

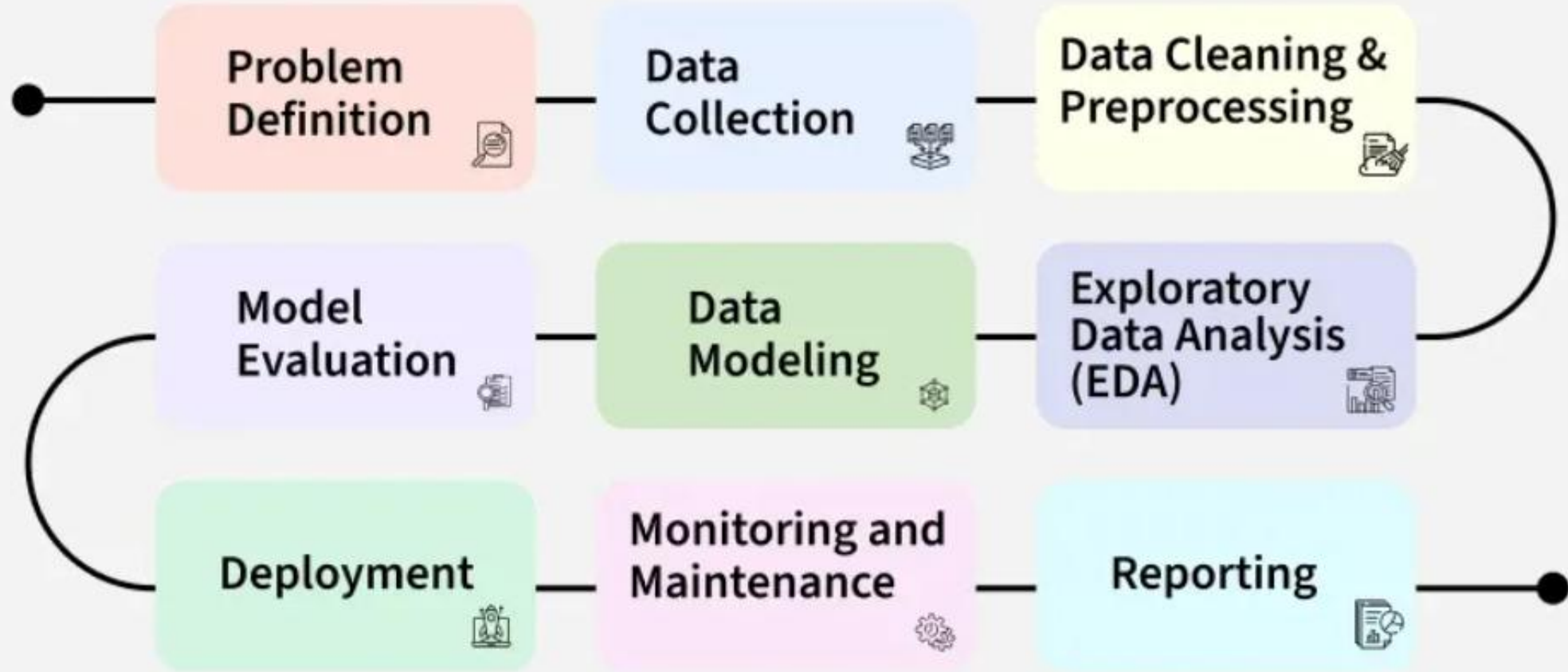
What's Data Science Pipeline?

- Data Science is a field that focuses on extracting knowledge from data sets that are huge in amount.
- It includes preparing data, doing analysis and presenting findings to make informed decisions in an organization.
- A pipeline in data science is a set of actions which changes the raw data from various sources to an understandable format so that we can store it and use it for analysis.

What's Data Science Pipeline?



What's Data Science Pipeline?



The raw data undergoes different stages within a pipeline which are:

- **Step 1: Problem Definition**

- The first step of Data Science Pipeline is understanding the problem we want to solve.
- This involves communicating with stakeholders to clearly define the objectives, the business or scientific questions and what success looks like.
- Having a well defined problem ensures that the data science efforts are focused and aligned with actual needs preventing wasted effort on irrelevant analysis.

Step 2: Data Collection

- After defining the problem the next step is gathering the data needed to address it.
- Data can come from various sources such as company databases, external APIs, sensors, public datasets or web scraping.
- Ensuring the data is relevant, sufficient and accessible is key and considerations like data privacy and compliance should be kept in mind during collection

Step 3: Data Cleaning and Preprocessing

- Data Cleaning involves cleaning and transforming the raw data into a usable format as it often contains errors, missing values, duplicates or inconsistencies.
- This involves handling missing or erroneous data, standardizing formats, encoding categorical variables, normalizing features and engineering new features that better represent the underlying problem.
- High quality preprocessing is important for building effective models.

Step 4: Exploratory Data Analysis (EDA)

- EDA is the process of exploring the cleaned data to understand its main characteristics through statistics and visualization. It helps uncover patterns, relationships, trends and anomalies.
- Techniques like plotting histograms, scatter plots and calculating correlations guide you to insights that influence feature engineering and model selection which helps refine hypotheses and the overall approach.

Step 5: Data Modeling

- Once the data is understood and prepared you choose suitable algorithms to build predictive or descriptive models.
- This includes splitting the data into training, validation and test sets, training models using various techniques like regression, classification or clustering and tuning hyperparameters to optimize performance.
- Model selection depends on the problem type and data characteristics.

Step 6: Model Evaluation

- After training models are evaluated using appropriate metrics such as accuracy, precision, recall, RMSE or AUC depending on the task which helps to verify how well the model generalizes to new data and whether it meets the success criteria.
- Evaluating performance helps detect issues like overfitting or underfitting, guiding decisions about further model refinement or feature adjustments.

Step 7: Deployment

- Successful models need to be put into production so they can provide value.
- Deployment involves integrating the model into existing systems via APIs, batch processes or real time streaming ensuring it runs efficiently and reliably at scale.
- This step often requires collaboration with engineering teams and consideration of infrastructure, security and maintenance needs.

Step 8: Monitoring and Maintenance

- Once deployed models need continuous monitoring to ensure they perform well over time which includes tracking prediction accuracy, detecting data drift or model decay and updating or retraining models as necessary.
- Maintenance guarantees that the model remains relevant and effective as new data and conditions evolve.

Step 9: Reporting

- Throughout and after the project it's important to communicate findings clearly to stakeholders which means creating understandable visualizations, reports and presentations that translate complex technical results into actionable business insights.
- Good communication ensures that the value created by data science is realized and drives informed decisions.

Thank You